

The Indus Script as a Mnemonic Framework: Ritual, Memory, and the Challenges of Decipherment

by: Avishai Roif

Edited and Abbreviated using an AI.

Abstract

The undeciphered script of the Indus Valley Civilization (2600–1900 BCE) remains one of archaeology's most enduring mysteries. This study argues that the Indus script functioned primarily as a mnemonic device, encoding ritual practices, proto-mantras, and ceremonial protocols through a hybrid system of pictographic symbolism and proto-linguistic notation. By analyzing recurring motifs (e.g., "unicorn," "horned deity," "chalice") within their archaeological contexts—fire altars, ritual baths, and burial sites—the paper posits that these symbols acted as ritual hyperlinks, triggering memorized chants or actions during religious ceremonies. Comparative evidence from Inca *quipu*, Egyptian hieroglyphs, and Vedic *sūtras* underscores the role of non-linguistic symbols in transmitting cultural knowledge. Computational methods, including AI-driven pattern recognition and 3D imaging, reveal structured sign clusters suggestive of metered recitations, while critiques of the script's brevity and entropy are addressed through hybrid functional models. The study integrates epigraphic, anthropological, and technological approaches to propose that the Indus script bridged oral and visual ritual traditions, offering a framework for future interdisciplinary decipherment efforts.

Introduction

The Indus Valley Civilization (2600–1900 BCE), one of the world's earliest urban societies, left behind a tantalizing enigma: thousands of inscribed seals, tablets, and pottery fragments bearing a script of over 400 unique symbols. Despite a century of scholarship, the script remains undeciphered, fueling debates over its purpose—linguistic, administrative, or ritual. A compelling hypothesis posits that the Indus script functioned as a mnemonic system, encoding ritual practices, proto-mantras, or ceremonial protocols rather than a full-fledged language. This essay evaluates the evidence for this theory, contextualizes it within broader archaeological and anthropological frameworks, and addresses critiques, while integrating recent advances in computational analysis.

Ritual and Memory in a Non-Literate Society

In pre-literate or semi-literate societies, visual symbols often serve as memory aids for transmitting complex cultural knowledge. The Indus script's structural features align with such systems:

Brevity of Inscriptions: Most texts average 4–5 signs, with rare exceptions exceeding 10 (Suthar, 2024). This contrasts with contemporaneous Mesopotamian cuneiform or Egyptian hieroglyphs, which encode extended narratives. The brevity suggests the script may have acted as shorthand for ritual sequences, akin to Vedic *sūtras*—concise formulae recited during ceremonies (Academia.edu, n.d.).

Pictographic Symbolism: Recurring motifs like the “unicorn” (often depicted alongside ritual objects) and the “horned deity” (interpreted as a proto-Shiva figure) likely held symbolic meanings tied to religious iconography (Kenoyer, 2010; Britannica, n.d.). Such symbols could have triggered memorized chants or actions during rituals, similar to how *mandalas* function in Hindu-Buddhist traditions. The “dual-circle” symbol, for instance, might have represented a cosmological duality principle, while a “tree-like motif” could have signified ancestry or divine connection (Preprints, 2025). The “chalice” motif found on some copper plates might have symbolized offering or divine communion, linking to early Mesopotamian and Hindu ceremonial objects (Preprints, 2025).

Archaeological Context: Inscriptions are concentrated in spaces linked to ritual activity—e.g., the Great Bath of Mohenjo-daro, fire altars at Kalibangan, and seals buried near gateways (Robbins Schug, 2020). This spatial correlation implies a ceremonial rather than administrative purpose (PWOnlyIAS, n.d.). The discovery of inscribed objects near fire altars and in ritual bathing areas like the Great Bath at Mohenjo-daro further supports this link.

Critics like Farmer and Weber (n.d.) argue that the script's non-linguistic nature precludes its use as a writing system and that the brevity of inscriptions limits their capacity for extensive mnemonic or accounting functions. However, mnemonic systems like the Inca *kipu* (knotted cords) or Aboriginal Australian songlines demonstrate that non-linguistic symbols can encode intricate knowledge (Smith, 2021; Cambridge, n.d.). The Indus script may have operated similarly, with symbols representing concepts, deities, or ritual actions rather than phonetic sounds.

Mnemonics in Ancient Cultures

The mnemonic hypothesis gains traction when compared to other systems:

Inca *Khipu*: Andean *kipus* used knots and colors to record numerical data and narratives. Though non-linguistic, they facilitated memory through tactile and visual cues (Smith, 2021). Similarly, Indus symbols like “fish” or “tree” might have denoted abstract concepts (e.g., fertility, water) through metaphorical associations (Parpola, n.d.).

Egyptian Hieroglyphs: Early hieroglyphs combined logographic and phonetic elements but retained ritual significance (Preprints, 2025). The Indus “horned deity” seal, for instance, may have symbolized a deity’s attributes, much like Anubis’ jackal-headed icon. The Indus script, like Egyptian hieroglyphs, might have combined ideographic and logographic systems (ARF Journals, 2024).

Vedic Mantras: The script’s repetitive sign clusters (e.g., “jar” + “fish”) mirror Vedic *mantras*, where phonetic patterns aid memorization (Academia.edu, n.d.). If the Indus signs represented ritual “keywords,” their repetition could reflect liturgical formulae. Some scholars propose linguistic connections between the Indus script and Vedic Sanskrit, suggesting potential use for sacred chants (digitalthought.info, n.d.). The “danda stick” marks in the Indus script might have functioned like repeat marks in music or indicated segments of verse and mantra (digitalthought.info, n.d.).

Challenges and Counterarguments

Lack of a Bilingual Text: Without a Rosetta Stone, decipherment relies on circumstantial evidence. Critics note that even non-linguistic systems like heraldry or traffic signs require cultural context for interpretation—context lost for the Indus Valley (Farmer & Weber, n.d.).

Statistical Ambiguity: Computational studies yield conflicting results. While some argue the script’s sign repetition matches linguistic patterns (Washington University, 2009), others highlight its low entropy (non-randomness), consistent with symbolic systems (Farmer & Weber, n.d.).

Overinterpretation of Symbols: The “unicorn” motif, for example, has been interpreted as a clan emblem, a zodiac symbol, and a ritual icon (Kenoyer, 2010; Digitalthought.info, n.d.). Such speculative leaps underscore the risks of projecting modern analogies onto ancient cultures.

AI and Interdisciplinary Approaches

Recent advances offer promising avenues:

Machine Learning: Projects like the 2025 *Indus Script Dictionary* use neural networks to map sign clusters against archaeological contexts (ResearchGate, 2025). Preliminary findings suggest recurring sequences near fire altars, possibly encoding offerings (e.g.,

“grain” + “fire” + “deity”). AI and machine learning are being applied to identify patterns and compare the Indus script with known languages (Times of India, 2024).

3D Imaging: High-resolution scans of seals have revealed previously unnoticed micro-symbols, hinting at layered meanings (Preprints, 2025).

Ethnographic Parallels: Comparing Indus symbols to modern Dravidian ritual art (e.g., *kolam* patterns) may uncover enduring mnemonic traditions (Times of India, 2024).

Toward a Hybrid Model

The Indus script likely served multiple functions—mnemonic, ritual, and proto-linguistic. As Suthar (2024) notes, “It may never have been meant to be ‘read’ but to be ‘remembered.’” While definitive decipherment remains elusive, the mnemonic hypothesis provides a robust framework for understanding how the Harappans bridged oral and visual culture. Future research must integrate computational tools, ethnographic insights, and targeted excavations to unravel this Bronze Age enigma.

Bibliography

Britannica. (n.d.). *Indus civilization*. Retrieved from <https://www.britannica.com/topic/Indus-civilization>

Farmer, S., & Weber, R. (n.d.). *The Collapse of the Indus-Script Thesis*. Retrieved from <https://safarmer.com/fsw2.pdf>

Kenoyer, J. M. (2010). *Master of Animals in the Indus Civilization*. Harappa.com. Retrieved from <https://www.harappa.com/sites/default/files/pdf/Kenoyer%202010%20Master%20of%20Animals.pdf>

Parpola, A. (n.d.). *Indus Script*. Retrieved from <https://www.harappa.com/sites/default/files/pdf/indusscript%282%29.pdf>

Preprints. (2025). *Deciphering the Indus Valley Script: AI-Driven Insights*. Retrieved from <https://www.preprints.org/manuscript/202502.0699/v1>

PWOnlyIAS. (n.d.). *The Script of the Indus Valley Civilisation Remains Undeciphered Despite Significant Archaeological Findings. Discuss The Challenges Faced In Decoding The Script And Suggest Measures To Promote Collaborative Research In This Field. (10 Marks, 150 Words) - PWOnlyIAS*. Retrieved from <https://pwonliyas.com/mains-answer-writing/the-script-of-the-indus-valley-civilisation-remains-undeciphered/>

ResearchGate. (2025). *Deciphering the Indus Valley Script: New Approaches and AI-Driven Insights Introduction to the Indus Valley Civilization and Its Script*. Retrieved from https://www.researchgate.net/publication/388609604_Deciphering_the_Indus_Valley_Script_New_Approaches_and_AI-Driven_Insights_Introduction_to_the_Indus_Valley_Civilization_and_Its_Script

Robbins Schug, G. (2020). Ritual. *Oxford Research Encyclopedia of Anthropology*.

Smith, M. (2021). *Indus Writing and Inka Khipu Compared*. UCLA Anthropology. Retrieved from <https://anthro.ucla.edu/wp-content/uploads/2020/08/2021-Smith-Indus-Writing-and-Inka-Khipu.pdf>

Suthar, D. (2024). The Indus Script: Challenges, Significance, and Government Role. *REFLECTIONS*, 2(1), 1–10.

Times of India. (2024). *4,000-year-old Indus script enigma waiting to be decoded: Will AI unlock the ancient mystery?* Retrieved from <https://timesofindia.indiatimes.com/science/4000-year-old-indus-script-enigma-waiting-to-be-decoded-will-ai-unlock-the-ancient-mystery/articleshow/118124829.cms>

Washington University. (2009). *Indus script encodes language, reveals new study of ancient symbols*. Retrieved from <https://www.washington.edu/news/2009/04/23/indus-script-encodes-language-reveals-new-study-of-ancient-symbols/>

Academia.edu. (n.d.). *The Indus Script as Ritual Shorthand, Mnemonic Symbolism and Proto-Mantras in a Ceremonial Context*. Retrieved from [https://www.academia.edu/127463906/The Indus Script as Ritual Shorthand Mnemonic Symbolism and Proto Mantras in a Ceremonial Context](https://www.academia.edu/127463906/The_Indus_Script_as_Ritual_Shorthand_Mnemonic_Symbolism_and_Proto_Mantras_in_a_Ceremonial_Context)

Parpola, A. (2015). *The Roots of Hinduism: The Early Aryans and the Indus Civilization*. Oxford University Press.

Kenoyer, J. M. (2010). *Master of Animals and the Renewal of Urban Ritual in the Indus Civilization*. In *Sindh Through the Centuries* (pp. 71–83)

Parpola, A. (1994). *Deciphering the Indus Script*. Cambridge University Press. Chapter 12: "The Dravidian Hypothesis" (pp. 166–189).

Parpola, A. (1994). *Deciphering the Indus Script*. Cambridge University Press. Chapter 5: "The Rebus Principle in the Indus Script" (pp. 102–135).

Parpola, A. (2015). *The Dravidian Solution to the Indus Script Problem*. In *Journal of Indo-European Studies*, 43(1–2), 257–276